Track D — Perception

RescueSwarm: work package and funding request

NIDAR 2026–27 · Track 1 · Mission 1

Ask: INR 76,245 · 1–2 students · Detector, tiling, geotagging, calibration, dataset

This track owns the 250 marks available for detection. It also owns the programme's most honest weakness: with the ground-truth apparatus and the field dataset both deferred, detection recall is a *modelled* quantity and there is currently no funded route to measuring it.

1 Scope and ownership

This track owns **detector, tiling, geotagging, calibration, dataset**. It is one of four delivery tracks defined in `docs/development-plan.md` §1.3; the integrated system argument lives in the master proposal, `docs/proposal/rescueswarm-proposal.pdf`, which this document does not restate.

2 What the detector actually receives

The sensor is a 12.3 MP module at 4056 × 3040 on a **1.55 µm** pixel pitch behind a 6 mm lens. The sizing chapter assumed a **1.82 µm** pitch — the same pixel *count* at a different pixel *size*, which means resolution on target is not what that chapter states.

At the 40 m search altitude, taking a supine adult as 1.7 m by 0.5 m:

Stage	GSD (cm/px)	Target (px^2)	COCO class
Sensor, native	1.03	13 600	large
After 2× downsample	2.07	3 400	medium
Sizing chapter, 60 m, 2×	3.65	640	**small**

The last row matters more than it looks. Published recall figures report average precision *by size class*, and small-object AP is routinely far below medium on the same model. Quoting the 47-pixel figure implicitly promises small-object performance the programme does not need to accept.

3 The inference budget, and where it does not close

Tiling is 640 px tiles at 20 % overlap. At 2× downsample that is 12 tiles per frame, and at 2 Hz, **24 inferences per second** — inside the accelerator's capability.

The temporal budget does not close. Requirement SYS-46 asks for at least 12 looks per target per pass so that multi-frame fusion has enough independent observations. At 40 m and 2 Hz the target is in frame for **7.9 looks** — a 34 % shortfall. Meeting 12 looks needs

$$r \geq \frac{n\,v}{D} = 3.06 \text{ Hz},$$

which is 37 inferences per second rather than 24. The trade is real and unresolved: raise the capture rate and pay about half again in inference load, fly slower, fly higher, or relax the look count. It should be decided deliberately rather than discovered.

The overlap geometry does hold: at 128 px of shared margin against an 82 px target, no survivor can be cut by a tile boundary without appearing whole in the neighbouring tile.

4 Geolocation: calibrate before fusing

The error budget makes an ordering argument that this track should follow literally. Random terms fall as $1/\sqrt{N}$ with frame count; systematic terms do not fall at all. Boresight calibration and a ground-plane height fix therefore matter *more* than additional frames, and fusion is worth having but not worth over-investing in — it moves the total from 0.75 m to 0.66 m, while calibration moves it from 4.57 m to 3.88 m before RTK is even applied.

Calibrate, then fuse. Not the other way round.

5 The honest position on recall

Detection recall is modelled and will remain modelled unless the deferred lines are restored. The ground-truth apparatus would have given detections something to be measured against; the field dataset would have given the detector Indian terrain to be trained on. Both were deferred at team direction to reach the requested figure.

This track should state that plainly in every review rather than presenting modelled recall as though it were measured. If any single deferred line is restored, this is the one with the largest effect on a scored outcome.

6 Funding request

Every figure below is generated from `docs/proposal/figures/track_budget.py`, which partitions the master competition budget. The four track asks plus the shared pool reconcile to the master figure exactly; that reconciliation is asserted in code and fails the build if it ever stops holding. **K** marks a line held at professional grade on purpose.

Item		**INR**
Camera + lens (1 x 11,100 x 3 ac)	**K**	33,300
AI accelerator (1 x 11,000 x 3 ac)	**K**	33,000
Subtotal		66,300
Customs duty and freight		0
GST		0
Contingency at 15%		9,945
TRACK D ASK		**76,245**

Deferred or institutionally held, and therefore not requested: Ground-truth apparatus, Indian SAR field dataset, Spare camera + lens, Training compute. These remain capabilities the track requires; they are simply not being bought. Where a deferral carries a technical consequence it is stated in the risk table below rather than left implicit.

7 Deliverables by phase

Phase	Dates	This track delivers
P1	24 Aug – 13 Sep	Camera and accelerator ordered. Survey-altitude position stated with its evidence.
P2	14 Sep – 4 Oct	Detector running on the accelerator at the budgeted rate. Boresight calibration procedure defined.
P3	5 Oct – 18 Oct	First airborne imagery at the flown altitude.
P5	9 Nov – 22 Nov	Detection and geotagging in the loop, end to end.

8 Risks owned by this track

Risk	Sev.	Mitigation
Recall is unmeasured	High	Ground truth and dataset both deferred. Mitigation is honesty in review plus opportunistic collection during flight tests; the real fix costs money and is a restoration decision.
Fusion look count fails SYS-46 at the flown altitude	High	7.9 looks against 12 required. Needs a decision on capture rate, groundspeed or altitude. Shared with Track C.
Fine-tuning at the wrong target scale	Medium	Altitude error changes pixels on target, so the detector should be fine-tuned across a band around 82 px rather than trained at a single scale.

9 Interfaces to other tracks

With	Dependency
Track A	Depends on a stable camera mounting plane. Boresight drift is a systematic error that no amount of frame fusion removes.
Track B	Depends on the accelerator, the camera interface, and on time and attitude being accurate enough that a geotag means something.
Track C	Supplies detections; consumes aircraft state. Shares the unresolved survey-altitude decision.

Interfaces are where student programmes fail, because each side assumes the other owns the gap. Every row above is a two-way commitment and should be recorded as an interface control document at P1.

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